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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	EyeScream
Selected Domain	Domain 1 – The Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4 – Quality Education (supporting: SDG 8 – Decent Work and Economic Growth)
Project / Solution Name	BridgeNext
Name (Leader)	Yap Li Shan
Email & Phone (Leader)	lishan0311@gmail.com · 016-272 9309
University Affiliation(s)	Asia Pacific University of Technology & Innovation (APU)
Number of Team Members	2

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Consider a financial operations assistant whose employer announces that invoice checking, reconciliation and reporting will move to AI-enabled workflows. They must decide whether to strengthen their current role or move to an adjacent career, with limited time, money and guidance on which skills will stay useful. BridgeNext focuses on this decision moment.

The problem is large in Malaysia. TalentCorp's impact study expects 620,000 jobs, 18% of employees in ten key sectors, to face significant changes from AI, digitalisation and the green economy within three to five years (Loheswar, 2024). Hays reports that 56% of working professionals in Malaysia use AI tools at work, but only 26% have received training (Hays, 2025). Globally, about one in four workers is in an occupation with some generative-AI exposure, with clerical work most exposed; exposure signals task change, not confirmed job loss (Gmyrek et al., 2025).

The economic risk is that changing task requirements weaken employability, while poorly targeted reskilling wastes scarce time and money. The social risk is unequal access to understandable career guidance and affordable learning, especially for workers who receive little employer support.

BridgeNext serves working adults adapting to AI-related change or considering a career move. The preliminary prototype focuses on financial operations and administrative support staff in Malaysia. It closes the gap between broad AI news and a person's next step through task-level AI evidence, transparent comparison of adjacent careers and time-budgeted learning pathways, while users keep control of every decision.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

BridgeNext is a deployed web application that helps working adults understand AI-related task changes, compare adjacent careers and plan affordable reskilling. It covers the six Domain 1 requirements:

1. Skill profile: A four-step questionnaire captures occupation, tools, tasks, skills, 1–5 confidence ratings, interests and weekly study time. Alternatively, users import a CV through browser-based PDF.js and Tesseract OCR. Free-text descriptions use catalogue keyword matching; users review suggestions before adding them.

2. Trend signals: Task-level AI-led and collaborative usage shares from the Anthropic Economic Index carry source, release and month labels. These describe sampled Claude usage, not Malaysian adoption rates. The backend discovers releases, validates compatible data and retains the last valid evidence when checks fail.

3. Explainable match: A rule-based score combines skill fit (70%), interests (20%) and scenario-adjusted fit (10%) across five adjacent directions, showing components, strengths and skill gaps.

4. Learning pathway: Ordered free resources include prerequisites, estimated study hours and schedules based on weekly learning time.

5. SDG mapping: Every recommended direction and skill includes an SDG target and an explanation.

6. Scenarios: "Today's work" and "more AI-led work" adjust routine and oversight skill priorities through explicit assumptions, not forecasts.

The preliminary prototype uses keyword matching and deterministic explanations. Optional Gemini integration code is prepared, but live activation and validation are deferred to Phase 2. Gemini is intended to suggest validated catalogue skills and explain computed results; deterministic rules retain control over scores and resources.

We apply Phase 1 workshop techniques in input validation, SQLite database design and cloud deployment. One Replit service hosts the React interface and FastAPI backend. Phase 2 will apply GenAI API integration, contextual prompting and structured JSON extraction. The interface supports English, Bahasa Malaysia and Simplified Chinese.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React 19, TypeScript and Vite; Recharts for the skill chart; Lucide icons; pdf.js and Tesseract.js for in-browser CV reading; English, Bahasa Malaysia and Simplified Chinese interfaces.

Backend: Python 3.12 with FastAPI, Pydantic and Uvicorn. One FastAPI service serves the built React app and the JSON API: /api/matches, /api/profile/extract, /api/evidence/refresh and /api/evidence/status.

Database: SQLite stores the reviewed catalogue (5 career directions, 18 skills, 6 task groups, 8 learning resources and their SDG mappings), dated task observations and an audit history of every evidence check.

AI: Gemini 2.5 Flash through Replit AI Integrations, requested as JSON at temperature 0. Returned skill IDs are checked against the catalogue before the user confirms them (Applicable in Phase 2).

Data: Anthropic Economic Index (26 June 2026 release; April and May 2026 task observations); O*NET OnLine occupation and task references; U.S. BLS 2025–35 projections, shown as context and excluded from scores; UN SDG targets. TalentCorp MyCOL and DOSM are linked as Malaysian context. Python scripts build the catalogue from these sources.

Matching: score = 70% baseline skill fit + 20% interest fit + 10% scenario-adjusted skill fit. AI-use evidence changes only the 10% component, through reviewed task-to-skill mappings. If any compared role has evidence for fewer than half of its linked tasks, the component is dropped for all roles and the other weights are renormalised; missing values are never treated as zero. Parity tests keep the browser and server engines identical.

Biggest risk and fallback: a new evidence release may change format or coverage. Each download is pinned to an immutable revision and validated (schema, periods, percentages, task coverage); if a check fails, the last valid evidence stays active and the failure is logged. If Gemini is unavailable, the app uses labelled catalogue keyword matching.

Deployment: code on public GitHub (Apache-2.0), deployed on Replit Deployments at bridge-next.replit.app; credentials stay in Replit Secrets. Appendix A shows the system block diagram.

Sponsor resources: Yes. Replit Core, provided for the hackathon, is used for development and hosting on Replit Deployments. The prototype has no GPU workload, so RunPod is not used.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: We researched Malaysian workforce data and a specific decision moment: a finance operations worker facing AI-enabled task changes. We have not interviewed users, so our choices remain research-informed.

Define: We focused the prototype on finance operations and administrative support, allowing manual review of roles, tasks and mappings.

Ideate: Limited study time informed weekly learning budgets; uncertainty informed two scenarios; trust informed visible sources, score components and "not assessed" labels.

Prototype: We deployed one service and refined the flow to show a recommended direction before optional comparisons. We removed ready-made personas so results reflect users' answers.

Test: Automated checks cover invalid inputs, missing evidence, an unconfigured model, refresh failures and browser–server consistency. By Phase 2, at least six target users will complete the same planning task using a static reference pack and BridgeNext, in counterbalanced order, assessed with one rubric. Deployment keeps credentials server-side and validates inputs; testing uses no real CVs.

Beyond the core Phase 2 scope, we plan optional peer-learning communities for sharing career-transition experiences, discussing resources and supporting shared learning goals.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary: SDG 4.4** – "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." BridgeNext identifies transferable skills and gaps and turns them into free, time-budgeted learning pathways. **Supporting: SDG 8.2** – higher economic productivity through diversification, technological upgrading and innovation. BridgeNext helps workers adapt to AI-enabled task change (United Nations, 2015). In the app, every direction and skill carries its own SDG statement; the sustainability specialist direction also maps to SDG 12.6. We measure planning capability and learning readiness, not national employment or productivity gains.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Published problem baseline:** 56% of working professionals in Malaysia use AI tools at work, but only 26% have received training (Hays, 2025). 620,000 jobs, 18% of employees in ten key sectors, are expected to face significant change within three to five years (Loheswar, 2024). About 25% of global employment is in occupations with some generative-AI exposure; this is not a Malaysian displacement rate (Gmyrek et al., 2025). **Product baseline:** task success without BridgeNext is not yet measured. Phase 2 testing will establish it with a fixed reference pack and the same rubric used for BridgeNext. Published training rates are not treated as product results.

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Primary KPI: career-planning task success rate (%).** Success requires a plausible career choice with a source-supported reason, one relevant skill gap, and a suitable learning resource and schedule within the stated time budget. Success rate = successful participants ÷ evaluated participants × 100. **Supporting KPIs:** correct identification of a skill that is a priority under both scenarios; source coverage of recommended roles, skills and resources; and affordable-resource coverage. These are project measures, not official UN indicators.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	By the Phase 2 presentation, evaluate at least six target users and aim for: 1. At least 80% planning-task success, above the reference-pack baseline. 2. At least 80% correctly identifying one skill that is a priority under both scenarios. 3. 100% source coverage for recommended roles, skills and resources. 4. 100% of recommended resources free or costing no more than RM100 each. We will report counts, percentages and the difference from baseline in percentage points; with six users, at least five must succeed (83.3%). Targets are proposed, not achieved results. In the current catalogue, all eight learning resources are free to access.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React 19 and React DOM; Vite; TypeScript	Code (UI framework, build tool, language)	MIT; MIT; Apache-2.0
Recharts; Lucide React	Code (chart and icon libraries)	MIT; ISC
PDF.js (pdfjs-dist); Tesseract.js with English OCR data	Code; pre-trained OCR model	Apache-2.0
FastAPI; Pydantic; Uvicorn	Code (backend framework, validation, server)	MIT; MIT; BSD-3-Clause
Python; SQLite	Code (language runtime; database engine)	PSF-2.0; public domain
google-genai SDK	Code (Gemini client)	Apache-2.0
Gemini 2.5 Flash via Replit AI Integrations — planned for Phase 2; not activated in the preliminary demonstration	Model (hosted, pre-trained; planned integration)	Google Gemini API and Replit terms
Anthropic Economic Index (26 June 2026 release)	Dataset	CC BY (data); MIT (code)
O*NET OnLine occupation and task references	Dataset	CC BY 4.0, attributed
U.S. BLS Table 1.2 (2025–35 projections)	Dataset	Public domain (U.S. Government)

UN SDG targets	Reference text	UN terms of use
TalentCorp MyCOL; DOSM	Linked references (not processed)	© TalentCorp; © DOSM
DM Sans and Manrope fonts	Font	SIL Open Font License 1.1

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature:  Yap Li Shan (Team Leader)

Date: 24 October 2026

References

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Appendix A. System Block Diagram

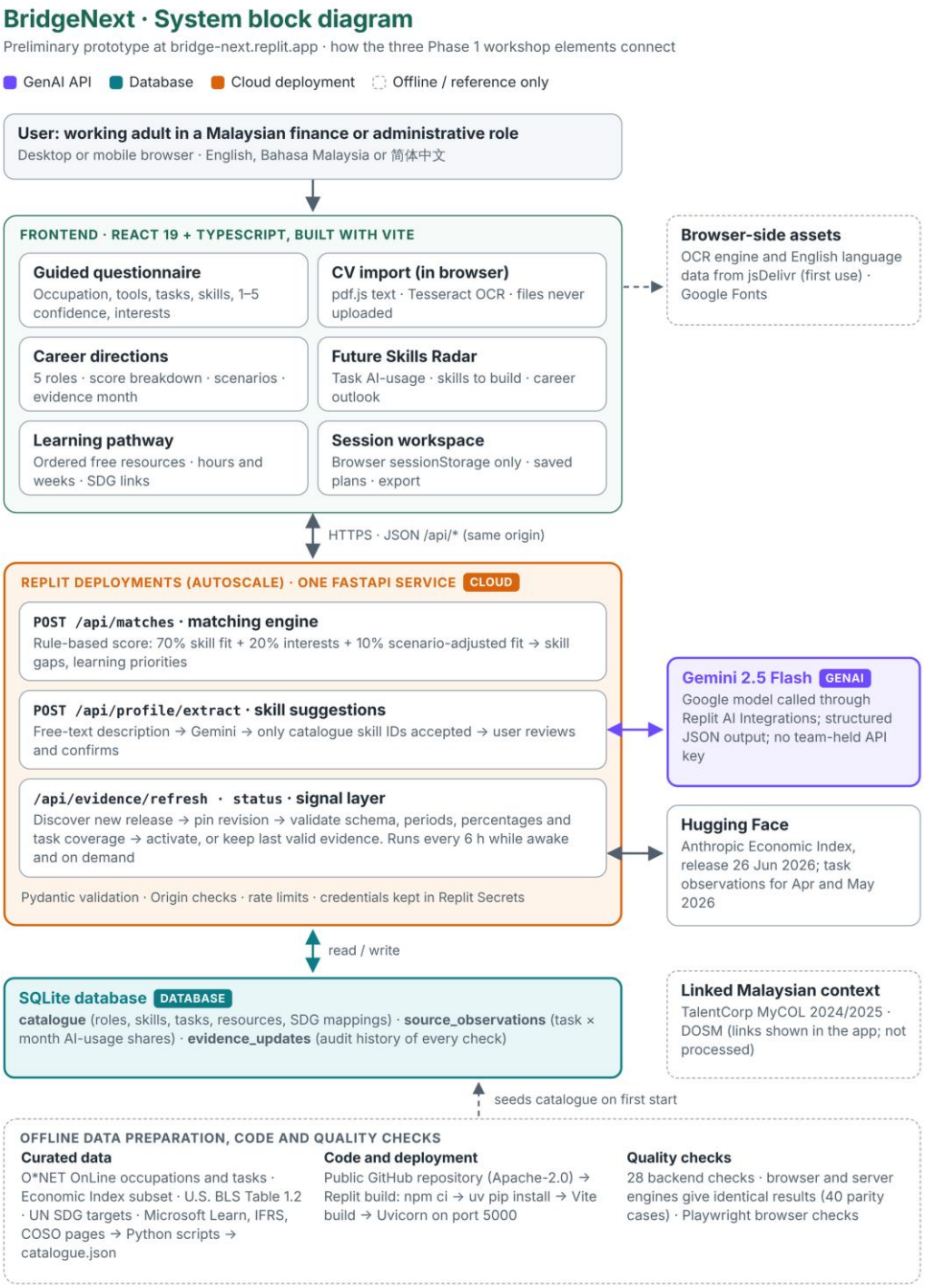


Figure A1. BridgeNext system architecture.